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Quality detection method and apparatus

Abstract

A method of fabricating a device involves forming a plurality of structures, such that each structure of the plurality includes a substrate and an epitaxial layer on the substrate. The epitaxial layer and the substrate have a lattice mismatch. The method further includes forming an electrical contact on the epitaxial layer of a selected structure of the plurality of structures and performing a current leakage measurement quality control test for the selected structure of the plurality of structures through the electrical contact. The method also involves forming a device on each of the remaining structures of the plurality of structures if the selected structure passed the leakage measurement quality control test or discarding each of the remaining structures of the plurality of structures if the selected structure did not pass the leakage measurement quality control test.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims priority to and the benefit of U.S. Provisional Application No. 63/214,541, filed Jun. 24, 2021, entitled “Highly Sensitive Apparatus and Method for Quality Detection in Epitaxial GaN” which is incorporated herein by reference in its entirety for all purposes.

BACKGROUND

(1) Wide-bandgap semiconductor materials, such as Gallium-Nitride (GaN) materials, have unique material characteristics, which includes low on-resistance, high operation frequency and high breakdown voltage. These benefits can make power conversion more energy and space efficient. GaN can be grown on silicon substrates, which allows the use of silicon manufacturing capability

and lower cost. However, GaN-on-Silicon have a substantial lattice mismatch, which often results in crystal defects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 illustrates the general methodology of the present disclosure.
- (3) FIG. 2A-B illustrates selecting a wafer with an epitaxial layer from a batch of wafers for leakage measurement.
- (4) FIG. 3A-D illustrates a process of forming a contact on an epitaxial layer of a wafer selected for leakage measurement.
- (5) FIGS. 4A-C and 5A-C provides comparison of quality control for the present methodology with a quality control procedure performed on a fully fabricated device with GaN epitaxial layer on a silicon substrate.
- (6) FIG. 6A-C illustrate possible modifications for wafers of a batch if a selected wafer of the batch passed the leakage test.
- (7) FIG. 7 illustrate an exemplary device, which can be formed on a wafer of a batch if another wafer of that batch passed the leakage test.
- (8) FIG. 8 shows an exemplary device with a seal ring region and a device region.
- (9) FIG. 9A-B show an exemplary device with a device region and a seal region surrounding a device region.
- (10) FIG. 10A-B show various stages of producing an exemplary device with a seal ring region and a device region.

DETAILED DESCRIPTION

(11) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(12) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(13) The present disclosure provide a methodology which involves a quality control test for a device, such as a semiconductor device, which involves forming on a substrate an epitaxial layer which has a lattice mismatch with the substrate, before forming additional components of the

device. The lattice mismatch may mean that a lattice parameter and/or a crystalline lattice structure of the formed epitaxial layer differ from the one or both of the substrate. Crystalline lattice structures may have one, two or three lattice parameters. The lattice mismatch may mean that at least one lattice parameter of a crystalline lattice structure of a material of the substrate differs from at least one lattice parameter of a crystalline lattice structure of the epitaxial layer by at least 0.5 Å, or at least 0.75 Å or at least 1.0 Å or at least 1.25 Å or at least 1.5 Å or at least 1.75 Å or at least 2.0 Å or at least 2.25 Å.

(14) FIG. 1 illustrates the present methodology. In step **110**, an epitaxial layer is formed on each substrate of a batch, which includes plural substrates, which can be overall identical. The epitaxial layer may be formed on each substrate of the batch in a single process. The formed epitaxial layer has a lattice mismatch with a material of the substrate(s). One substrate of the batch with the formed epitaxial layer, may be selected for a quality control test. In step **130**, an electrical ohmic contact, such as a metallic contact, may be formed on the epitaxial layer, of the substrate selected for the quality control test. The electrical ohmic contact may have lateral dimensions, i.e. dimensions parallel to the surface of the substrate, which correspond to a size of an active region for a desired semiconductor device. In step **150**, the quality control test is performed on the selected substrate through the electrical contact. The quality control test involves applying a voltage between the electrical contact and the bottom surface of the substrate and measuring a current leakage on between the epitaxial layer and the substrate. If the selected substrate fails the quality control test, e.g. the measured leakage current is greater than a threshold leakage current value, then all the substrates of the batch with their epitaxial layers are discarded, see step **190**. If the selected substrate passes the quality control test, e.g. the measured leakage current does not exceed the threshold leakage current value, then all the remaining substrates of the batch with their epitaxial layers, are used for forming a device, such as a semiconductor device, see step **190**. In some embodiments, the selected substrate, which passed the quality control test, may be discarded after the quality control test. Yet in some embodiments, a semiconductor device may be fabricated on the selected substrate, which passed the quality control test. Such semiconductor device may be a semiconductor device with a seal ring region shown in FIGS. 8 and 9A-B. The ohmic contact on the selected substrate, which passed the quality control test, may become a part of the seal ring region.

(15) FIG. 2A schematically illustrates a batch of substrates **200A-200H**. Each of these substrates has an epitaxial layer formed on the material of the substrate in a single process. The material of the substrates and the epitaxial layer have a lattice mismatch. As shown in FIG. 2A, each of the substrates **200A-200H** may be placed on a holder **201** and introduced in a deposition chamber, such as an epitaxial growth chamber, in which the epitaxial layer may be formed on each of the substrate **200A-200H** simultaneously. One of the substrates **200A-200H**, substrate **200H** in FIG. 2A is selected from a quality control leakage measurement. FIG. 2B shows substrate **200H** with original substrate material **202** of the substrate with epitaxial layer **204**, formed on original substrate material **202**, and ohmic electrical contact element **206**, such as a metal contact, on top of epitaxial layer **204**. Electrical contact **206** may have optional insulating material **208** along its side edges. For the quality control test, an electrode **210** of a quality testing apparatus is brought into electrical contact with ohmic electrical contact element **206** to provide electrical voltage between electrical contact element **206** and bottom of original substrate material **202**. The quality testing apparatus measures a current leakage between original substrate material **202** and epitaxial layer **204**. If the measured current leakage for substrate **200H** is above a current leakage threshold value, then each of remaining substrates **200A-200G** of the batch is discarded. If the measured current leakage for substrate **200H** does not exceed the current leakage threshold value, then each of remaining substrates **200A-200G** (each of which include an original substrate material layer **200** similar to layer **202** of tested substrate **200H** and an epitaxial layer, similar to layer **204** of tested substrate **200H**) is used for forming a device, such as semiconductor device. Lateral dimensions of ohmic

electrical contact **206** in substrate **200H** selected for the current leakage quality control test may correspond to lateral dimensions of an active region of the device, such as a region defined by shallow trench isolation, formed on each of the remaining substrates **200A-200G** of the batch. In some embodiments, one or both lateral dimensions of ohmic electrical contact **206** may be 1 micron or more.

(16) Although FIG. 2A illustrates the batch, which has eight substrates, a batch size may vary. In some embodiments, a batch may include at least 5 substrates or at least 6 substrates or at least 7 substrates or at least 8 substrates or at least 9 substrates or at least 10 substrates or at least 11 substrates or at least 12 substrates or at least 13 substrates or at least 14 substrates or at least 15 substrates or at least 16 substrates or at least 17 substrates or at least 18 substrates or at least 19 substrates or at least 20 substrates.

(17) FIG. 3A-D illustrate forming ohmic electrical contact element **206** on epitaxial layer **204**, of substrate **200H** selected for the quality control leakage test. For the sake of simplicity, FIG. 3A-D do not show original substrate material layer **202** underneath epitaxial layer **204**. In FIG. 3A, electrically conducting material **206** is deposited on the top surface of epitaxial layer **204**. The electrically conducting material **206** may be for example, electrically conducting carbon, or a metal, such as copper, titanium, tungsten, aluminum, silver, gold, ruthenium, platinum, or their alloys. In some embodiments, the electrically conducting material **206** may be one or more of titanium, aluminum, AlCu or TiN. The electrically conducting material **206** may be deposited using a physical deposition technique, such as evaporation or sputtering. In FIG. 3B, electrically conducting material **206** is patterned using a mask. The mask may provide electrically conducting material **206** with lateral dimensions which may correspond to lateral dimensions of an active region of a device, which is considered to be formed on the remaining substrates of the batch, such as substrates **200A-200G**, if selected substrate **200H** passes current leakage quality control test. As the result of the patterning, the lateral dimensions of the electrically conducting material **206** are reduced and a portion of the top surface of epitaxial layer **204** is reexposed. In FIG. 3C, insulating layer **208** is deposited over the patterned electrically conducting material **206**. Insulating layer **208** may be an oxide layer, such as a silicon oxide layer. Insulating layer **208** may be formed using for example, a chemical vapor deposition. Insulating layer **208** may be implanted with oxygen atoms and annealed. The annealing process may be rapid thermal processing performed at a temperature from 400 C to 500 C, such as 450 C, for a few seconds, such as 30 seconds or less, 20 seconds or less, 15 seconds or less, 10 seconds or less, 5 seconds or less. Insulating layer **208** may protect surface morphology of epitaxial layer **206** during implantation with oxygen atoms and annealing. In FIG. 3D, most of insulating layer **208** may be etched to expose top surface of patterned electrically conducting material **206**. Some portions of insulating layer **208** may remain of outer sides of patterned electrically conducting material **206**. Insulating layer **208** may be etched using a dry etching technique, such as plasma etching, a wet etching technique, which may be, for example, buffered HF etching, or their combination. The substrate **200H** after the modifications as showed in FIG. 3D is subjected for the current leakage quality test as shown in FIG. 2B.

(18) The substrate is formed of a material, such as a crystalline material, which has a lattice mismatch, with a subsequently grown epitaxial layer.

(19) In some embodiments, the material of the substrate may be a semiconductor material, such as a Group IV semiconductor material, a Group III-V semiconductor material or a Group II-VI semiconductor material. The semiconductor material of the substrate may be doped or undoped.

(20) Examples of Group IV semiconductor materials which may be used as a substrate include elemental Group IV crystalline semiconductor materials, such as crystalline silicon, crystalline germanium or crystalline tin; and binary Group IV crystalline semiconductor materials, such as SiGe, SiC or GeC; ternary Group IV crystalline semiconductor materials, such as SiGeC. Group IV semiconductor materials may be n-doped with one or more n-dopants, such as phosphorous, antimony, arsenic, bismuth or lithium. Group IV semiconductor materials may p-doped with one or

more p-dopants, such as boron, aluminum, gallium or indium.

(21) Examples of Group III-V semiconductor materials which may be used as a substrate include binary Group III-V crystalline semiconductor materials, such as BN, AlN, GaN, InN, BP, AlP, GaP, InP, BAs, AlAs, GaAs, InAs, BSb, AlSb, GaSb, InSb; and ternary Group III-V crystalline semiconductor materials, such as $\text{In.sub.xGa.sub.1-xN}$ ($0 < x < 1$), $\text{In.sub.xGa.sub.1-xP}$ ($0 < x < 1$), $\text{In.sub.xGa.sub.1-xAs}$ ($0 < x < 1$), $\text{In.sub.xGa.sub.1-xSb}$ ($0 < x < 1$), $\text{Al.sub.xIn.sub.1-xN}$ ($0 < x < 1$), $\text{Al.sub.xIn.sub.1-xP}$ ($0 < x < 1$), $\text{Al.sub.xIn.sub.1-xAs}$ ($0 < x < 1$), $\text{Al.sub.xIn.sub.1-xSb}$ ($0 < x < 1$), $\text{Al.sub.xGa.sub.1-xN}$ ($0 < x < 1$), $\text{Al.sub.xGa.sub.1-xP}$ ($0 < x < 1$), $\text{Al.sub.xGa.sub.1-xNAs}$ ($0 < x < 1$), $\text{Al.sub.xGa.sub.1-xSb}$ ($0 < x < 1$). Group III-V semiconductor materials may be doped with one or more n-dopants, such as tellurium, sulfur (substituting Group V element(s)); tin, silicon, germanium (substituting Group III element(s)). Group III-V semiconductor materials may be doped with one or more p-dopants, such as beryllium, zinc, chromium (substituting Group III element(s)); silicon, germanium, carbon (substituting Group V element(s)).

(22) Examples of Group II-VI semiconductor materials include CdTe and CdS which may be used as a substrate. Group II-VI semiconductor materials may be doped with one or more n-dopants, such as a Group III element, such as Al, Ga, In, substituting the Group II element(s), such as Cd; or a halogen, such as F, Cl, I or Br, substituting the Group VI element(s). Group II-VI semiconductor materials may be doped with one or more p-dopants, such as a Group V element, such as P, substituting the Group VI; or a group I element, such as lithium or sodium, substituting the Group II element.

(23) In some embodiments, the material of the substrate may be a non-semiconductor crystalline material, such as an insulating crystalline material, such as sapphire or quartz.

(24) In some embodiments, the epitaxial layer, may comprise at least one crystalline semiconductor material, such as a Group IV crystalline semiconductor material, a Group III-V crystalline semiconductor material or a Group II-VI crystalline semiconductor material.

(25) Examples of Group IV semiconductor materials which may be used in the epitaxial layer, include elemental Group IV crystalline semiconductor materials, such as crystalline silicon, crystalline germanium or crystalline tin; and binary Group IV crystalline semiconductor materials, such as SiGe, SiC or GeC; ternary Group IV crystalline semiconductor materials, such as SiGeC. Group IV semiconductor materials may be n-doped with one or more n-dopants, such as phosphorous, antimony, arsenic, bismuth or lithium. Group IV semiconductor materials may p-doped with one or more p-dopants, such as boron, aluminum, gallium or indium.

(26) Examples of Group III-V semiconductor materials which may be used in the epitaxial layer include binary Group III-V crystalline semiconductor materials, such as BN, AlN, GaN, InN, BP, AlP, GaP, InP, BAs, AlAs, GaAs, InAs, BSb, AlSb, GaSb, InSb; and ternary Group III-V crystalline semiconductor materials, such as $\text{In.sub.xGa.sub.1-xN}$ ($0 < x < 1$), $\text{In.sub.xGa.sub.1-xP}$ ($0 < x < 1$), $\text{In.sub.xGa.sub.1-xAs}$ ($0 < x < 1$), $\text{In.sub.xGa.sub.1-xSb}$ ($0 < x < 1$), $\text{Al.sub.xIn.sub.1-xN}$ ($0 < x < 1$), $\text{Al.sub.xIn.sub.1-xP}$ ($0 < x < 1$), $\text{Al.sub.xIn.sub.1-xAs}$ ($0 < x < 1$), $\text{Al.sub.xIn.sub.1-xSb}$ ($0 < x < 1$), $\text{Al.sub.xGa.sub.1-xN}$ ($0 < x < 1$), $\text{Al.sub.xGa.sub.1-xP}$ ($0 < x < 1$), $\text{Al.sub.xGa.sub.1-xNAs}$ ($0 < x < 1$), $\text{Al.sub.xGa.sub.1-xSb}$ ($0 < x < 1$). Group III-V semiconductor materials may be doped with one or more n-dopants, such as tellurium, sulfur (substituting Group V element(s)); tin, silicon, germanium (substituting Group III element(s)). Group III-V semiconductor materials may be doped with one or more p-dopants, such as beryllium, zinc, chromium (substituting Group III element(s)); silicon, germanium, carbon (substituting Group V element(s)).

(27) Examples of Group II-VI semiconductor materials include CdTe and CdS which may be used in the epitaxial layer. Group II-VI semiconductor materials may be doped with one or more n-dopants, such as a Group III element, such as Al, Ga, In, substituting the Group II element(s), such as Cd; or a halogen, such as F, Cl, I or Br, substituting the Group VI element(s). Group II-VI semiconductor materials may be doped with one or more p-dopants, such as a Group V element, such as P, substituting the Group VI; or a group I element, such as lithium or sodium, substituting

the Group II element.

(28) In some embodiments, the epitaxial layer may be a single layer film of a material having a lattice mismatch with the material of the substrate. Yet in some embodiments, the epitaxial layer may be a multilayer film formed of subsequently epitaxially grown layers, such that two adjacent layers have different compositions. At least one epitaxially grown layer of the multilayer film has a lattice mismatch with the material of the substrate.

(29) In some embodiments, the epitaxial layer may be grown directly on the material of the substrate. Yet in some embodiments, the epitaxial layer may be grown on a buffer layer formed directly on the material of the substrate. The buffer layer may be formed of a crystalline material having a lattice parameter between those of the material of the substrate and the material of the epitaxial layer. The buffer layer may be used to reduce the effect of the lattice mismatch between the material of the substrate and the material of the epitaxial layer.

(30) Table 1 provides lattice parameters and crystalline structures for selected materials, which may be used as a material of the substrate, a material of the epitaxial layer or both.

(31) TABLE-US-00001 TABLE 1 Lattice Constants and Crystal structures of selected materials.

Material	Lattice constant (Å)	Crystal structure
C (diamond)	3.567	Diamond (FCC)
C (graphite)	a = 2.461 Hexagonal c = 6.708	Graphite (FCC)
Si	5.431020511	Diamond (FCC)
Ge	5.658	Diamond (FCC)
AlAs	5.6605	Zinc blende (FCC)
AlP	5.4510	Zinc blende (FCC)
AlSb	6.1355	Zinc blende (FCC)
GaP	5.4505	Zinc blende (FCC)
GaAs	5.653	Zinc blende (FCC)
GaSb	6.0959	Zinc blende (FCC)
InP	5.869	Zinc blende (FCC)
InAs	6.0583	Zinc blende (FCC)
InSb	6.479	Zinc blende (FCC)
MgO	4.212	Halite (FCC)
SiC	a = 3.086 Wurtzite c = 10.053	Wurtzite (HCP)
CdS	5.8320	Zinc blende (FCC)
CdSe	6.050	Zinc blende (FCC)
CdTe	6.482	Zinc blende (FCC)
ZnO	a = 3.25 Wurtzite (HCP) c = 5.2	Wurtzite (HCP)
ZnS	4.580	Halite (FCC)
PbS	5.9362	Halite (FCC)
PbTe	6.4620	Halite (FCC)
BN	3.6150	Zinc blende (FCC)
BP	4.5380	Zinc blende (FCC)
CdS	a = 4.160 Wurtzite c = 6.756	Wurtzite c = 6.756
ZnS	a = 3.82 Wurtzite c = 6.26	Wurtzite c = 6.26
AlN	a = 3.112 Wurtzite c = 4.982	Wurtzite c = 4.982
GaN	a = 3.189 Wurtzite c = 5.185	Wurtzite c = 5.185
InN	a = 3.533 Wurtzite c = 5.693	Wurtzite c = 5.693
A-Al.sub.2O.sub.3	Sapphire a = 4.785 Hexagonal c = 12.991	Hexagonal c = 12.991
SiO.sub.2 α-quartz	a = 4.9134 Trigonal c = 5.4052	Trigonal c = 5.4052

(32) In some embodiments, the substrate may be a Group IV semiconductor substrate, such as a crystalline silicon substrate and the epitaxial layer may comprise a Group III-V semiconductor material, such as BN, AlN, GaN, InN, BP, AlP, GaP, InP, BAs, AlAs, GaAs, InAs, BSb, AlSb, GaSb, InSb, In.sub.xGa.sub.1-xN (0<x<1), In.sub.xGa.sub.1-xP (0<x<1), In.sub.xGa.sub.1-xAs (0<x<1), In.sub.xGa.sub.1-xSb (0<x<1), Al.sub.xIn.sub.1-xN (0<x<1), Al.sub.xIn.sub.1-xP (0<x<1), Al.sub.xIn.sub.1-xAs (0<x<1), Al.sub.xIn.sub.1-xSb (0<x<1), Al.sub.xGa.sub.1-xN (0<x<1), Al.sub.xGa.sub.1-xP (0<x<1), Al.sub.xGa.sub.1-xNAs (0<x<1), or Al.sub.xGa.sub.1-xSb (0<x<1).

(33) In some embodiments, the substrate may be an insulating crystalline substrate, such as a sapphire substrate or a quartz substrate and the epitaxial layer may comprise a Group III-V semiconductor material, such as BN, AlN, GaN, InN, BP, AlP, GaP, InP, BAs, AlAs, GaAs, InAs, BSb, AlSb, GaSb, InSb, In.sub.xGa.sub.1-xN (0<x<1), In.sub.xGa.sub.1-xP (0<x<1), In.sub.xGa.sub.1-xAs (0<x<1), In.sub.xGa.sub.1-xSb (0<x<1), Al.sub.xIn.sub.1-xN (0<x<1), Al.sub.xIn.sub.1-xP (0<x<1), Al.sub.xIn.sub.1-xAs (0<x<1), Al.sub.xIn.sub.1-xSb (0<x<1), Al.sub.xGa.sub.1-xN (0<x<1), Al.sub.xGa.sub.1-xP (0<x<1), Al.sub.xGa.sub.1-xNAs (0<x<1), or Al.sub.xGa.sub.1-xSb (0<x<1).

(34) The leakage current quality control test, such as the one illustrated in FIG. 2B, involves applying a voltage between the electrical contact **206** of epitaxial layer **204** and the bottom of substrate **202**. For applying the voltage, the bottom of substrate **202** may be grounded. The applied voltage may depend on specific materials of substrate **202** and epitaxial layer **204**. In some embodiments, the applied voltage may sweep, for example, from 0V to 1500V. A current-voltage slope for the applied voltage in the leakage current quality control test may be, for example, from 0.1 nA/V to 100 mA/V. A threshold leakage current may depend of a surface area of ohmic

electrical contact **206** parallel to the surface of substrate **202** and/or the surface of epitaxial layer **204**. In other words, the threshold leakage current may depend on the lateral dimensions of ohmic electrical contact **206**, which may correspond to lateral dimensions of an active region of a semiconductor device, which will be formed on the remaining substrates of the batch, such as substrates **200A-200G**, if the selected substrate of the batch, such as substrate **200G** passes the leakage current quality control test. In some embodiments, the threshold leakage current may be from 0.2 nA to 500 μ A. In some embodiments, the threshold current leakage value was about 10 mA.

(35) FIGS. **4A-C** and **5A-C** provide comparison of quality control for the present methodology with a quality control procedure performed on a fully fabricated device with GaN epitaxial layer on a silicon substrate.

(36) FIGS. **4A** and **5A** show an exemplary semiconductor device which was formed on each of seven substrates of an eight substrate batch, such the one shown in FIG. **2A**. The eight substrate of the same batch was used for the current leakage quality control test was used. Each of the eight substrates of the batch had GaN epitaxial layer **204** formed on silicon substrate material **202**. The device of FIGS. **4A** and **5A** has source **216**, drain **218** and gate **217** formed on epitaxial layer **204** grown on original substrate material **202**. The device of FIGS. **4A** and **5A** also has metallization contacts **219** formed on source **216** and drain **218**. FIGS. **4B** and **5B** provide comparison between a quality control test performed to the seven substrates with the fully fabricated device shown in FIGS. **4A** and **5A** and the current leakage quality control test according to the present methodology, which uses only a single substrate of the batch with the ohmic electrical contact **206** as shown in FIGS. **4C** and **5C** (essentially identical to FIG. **2B**). FIG. **4B** is a plot of a critical parameter (CP) for quality control measured on the fully fabricated device (FIG. **4A**) though metallization contacts **219** on drain **218** for the seven substrates of the batch versus a yield determined using the current leakage quality control test according to the present methodology performed on the single substrate of the batch before fabricating a full device as shown in FIG. **4C**. FIG. **4B** shows a correlation between the critical parameter (CP) for quality control measured on the fully fabricated device (FIG. **4A**) though metallization contacts **219** on drain **218** for the seven substrates of the batch versus the yield determined using the current leakage quality control test according to the present methodology performed on the single substrate of the batch before fabricating a full device as shown in FIG. **4C**.

(37) FIG. **5B** reports yields for multiple (eight) substrate batches determined (a) by testing each of fully fabricated devices (FIG. **5A**) though metallization contacts **219** on drain **218** on each of the seven substrates of the batch (the curve with crosses) and (b) determined using the current leakage quality control test according to the present methodology performed on the single substrate of the batch before fabricating a full device as shown in FIG. **5C** (the curve with rhomb s).

(38) The data in FIGS. **4B** and **5B** indicate that the present methodology can predict a yield and a quality of a multi-substrate batch without performing a time consuming process, which may take several weeks, such as e.g. eight weeks, for fabrication a full device, while performing the current leakage quality control test only on one substrate of the test. The present methodology may improve yield for batches of substrates, which have epitaxial layer(s) having a lattice mismatch with a material of the substrate and/or reduce testing time and/or process cycle time for semiconductor devices formed on substrates of such batches. The present methodology may provide one or more of the following advantages: (a) it may provide a guarantee for quality of substrates of a batch with epitaxial layer(s) of a material of the substrate; (b) it may save process and/or testing costs because for batches, in which a selected substrate fails the current leakage quality control test, a time consuming device fabrication process will not be performed; (c) it may significantly reduce a cycle time for yield improvement for devices fabricated on substrates having epitaxial layer(s) such as from several weeks, e.g. 8 weeks to 1 day; (d) the process for preparing the selected substrate to the current leakage quality control test is a low cost, one metal process,

which does not use an additional mask; (e) all the substrates of the batch that passed the current leakage quality control test for the selected substrate may be trackable.

(39) For devices with crystalline silicon as original substrate material **202** and GaN epitaxial layer **204**, currently existing quality control methodology are limited to in-line optical measurements. However, the in-line optical measurements do not provide a reliable prediction for the quality of the final device after the time consuming process for fabricating a device, such as the one in FIGS. **4A** and **5A**.

(40) FIGS. **6A-C** schematically illustrate operations, which may be performed on each remaining substrate of the batch, such as each of substrates **200A-200G** in FIG. **2A**, if the selected substrate, such as substrate **200H**, passes the current leakage quality control test. These operations may include forming shallow trench isolation structure **211** in epitaxial layer **204**. FIG. **6A** shows forming an insulating layer **212** over epitaxial layer **204** defined by shallow trench isolation structure **211**. FIG. **6B** shows forming metal layer **213** over epitaxial layer **204** defined by shallow trench isolation structure **211**. FIG. **6C** shows forming semiconductor layer **214** over epitaxial layer **204** defined by shallow trench isolation structure **211** and then forming ohmic contact layer **215** on semiconductor layer **214**. Lateral dimensions of each layers **212**, **213**, **214** and **215** may correspond to those of lateral dimensions of ohmic electrical contact **206** of the selected substrate, such as substrate **200H**, which was selected for the current leakage quality control test.

(41) Although FIGS. **4A** and **5A** shows only a single transistor for the sake of simplicity, in many embodiments, plural transistors may be formed on epitaxial layer **204**.

(42) FIG. **7** shows a device similar to the device shown in FIGS. **4A** and **5A**. However, epitaxial layer **204** in FIG. **7** is a multilayer formed on multiple layers **204A**, **204B**, **204C**, **204D** and **204E**. Each layer of the multilayer may differ an adjacent layer of the multilayer in at least one of a composition, doping and crystal structure. At least one layer of the multilayer is an epitaxial layer. In some embodiments, each of the layers of the multilayer may be an epitaxial layer. At least one epitaxial layer of the multilayer may have a lattice mismatch with a material of substrate **202**.

(43) The present disclosure also provides a semiconductor device which has a substrate and an epitaxial layer on the substrate. The epitaxial layer and the substrate have a lattice mismatch. The device includes a device region on a first part of a surface of the epitaxial layer and a seal ring region on a second part of the surface of the epitaxial layer. The device region may include a plurality of transistors; while the seal ring region may include an ohmic contact on the epitaxial layer. The ohmic contact may be used for performing a leakage current measurement on the final semiconductor device or at an intermediate stage of fabricating the device. The leakage current measurement may be used as a quality control test. The substrate with the epitaxial layer may be, for example, one of substrates **200A-200G** in FIG. **2A**. However, the semiconductor device does not necessarily is used together with the leakage current measurement test on selected substrate **200H** as illustrated in FIG. **2B**.

(44) FIG. **8** shows exemplary semiconductor device **230** which has a device region and a seal ring region. Device **230** has substrate **202** and epitaxial layer **204** on substrate **202**. Epitaxial layer **204** and substrate **202** have a lattice mismatch. Device **230** includes device region **222** on a first part of a surface of epitaxial layer **204** and seal ring region **224** on a second part of the surface of epitaxial layer **204**. Device region **222** includes region **221** with a plurality of transistors. Each of transistors may include a source, a drain and a gate, such as elements **216**, **217** and **218** in FIGS. **4A** and **4B**. Seal ring region **224** includes ohmic contact **223** on epitaxial layer **204**. Device **230** includes one or more insulating/passivation layers shown for illustration purposes only as element **226**. Such insulating/passivation layers may be formed of one or more of silicon oxide, silicon nitride or a combination thereof. Device **230** includes metallization contacts **219** (M1 closest to epitaxial layer **204** and Mn furthest from epitaxial layer **204**) extending through one or more insulating/passivation layers **226**. Metallization contacts **219** contact are provided for both seal ring

region **224** so that M1 has electrical contact with ohmic contact **223** and for device region **222** so that M1 has electrical contact with a drain of one or more transistors in region **221**. For device region **222**, top contact(s) Mn are exposed/open through opening(s) **225**. In some embodiments, for seal region **224**, top contact(s) Mn may be exposed/open through opening(s) **227**. Yet in some embodiments, top contact(s) Mn may be covered by top insulating/passivating layer **226**. The top passivating/insulating layer may be formed of one or more of silicon oxide, silicon nitride or a combination thereof. Ohmic contact **223** may be similar to ohmic contact **206** in FIGS. **4C** and **5C**. Ohmic contact **223** may have each of its lateral dimensions, i.e. dimensions parallel to the surface of epitaxial layer **204**, greater than 1 micron. Ohmic contact **223** may be formed by the process similar to the process for forming ohmic contact **206** in FIGS. **3A-D**. Such process has a short process time, a good contrast and provides a good contact between metal of the ohmic contact and the epitaxial layer **204**.

(45) FIGS. **9A-B** show device **230'** where seal ring region **224** surrounds device region **222**, see FIG. **9A** which shows cross section perpendicular to the plane of FIG. **9A**. In FIG. **9B**, top contact(s) Mn for seal region **224** are not exposed/open. Instead, top contact(s) Mn for seal region **224** are covered by top insulating/passivating layer **226**.

(46) When top contact(s) Mn for seal region **224** are exposed/open, ohmic contact **223** may be used for current leakage measurement. For such current leakage measurement, a configuration similar to the one in FIGS. **4C** and **5C** may be used. For the current leakage measurement, a voltage may be applied between top contact Mn in seal ring region **224** and the bottom of substrate **202** through ohmic contact **223** on epitaxial layer **204**. For applying the voltage, the bottom of substrate **202** may be grounded. The current leakage measurement performed through ohmic contact **223** on epitaxial layer **204** may serve as a quality control measurement similarly to the measurement in FIGS. **4C** and **5C**. The ability of performing a quality control measurement through ohmic contact **223** on epitaxial layer **204** in seal ring region **224** allows keeping device region **222**, including its top contacts Mn, in an original, untouched state.

(47) Ohmic contact **223** on epitaxial layer **204** may also be used for current leakage measurement at an intermediate stage of fabricating a semiconductor device having a seal ring region and a device region, such as devices **230** and **230'** in FIGS. **8** and **9**. FIG. **10A** shows a device, such as device **230**, at an intermediate stage after ohmic contact **223** and region **221** with a plurality of transistors, are formed on epitaxial layer **204**. FIG. **10B** shows a device, such as device **230**, at intermediate stage after lowest metallization contacts M1 are fabricated. In seal region **224**, M1 has electrical contact with ohmic contact **223**. In device region **222**, M1 has electrical contact with a drain of one or more transistors in region **221**. For current leakage measurement at the intermediate stage illustrated in FIG. **10A**, a voltage may be applied between ohmic contact **223** on epitaxial layer **204** and the bottom of substrate **202**. For applying the voltage, the bottom of substrate **202** may be grounded. For current leakage measurement at the intermediate stage illustrated in FIG. **10B**, a voltage may be applied between contact M1 in seal ring region **224** and the bottom of substrate **202** through ohmic contact **223** on epitaxial layer **204**. For applying the voltage, the bottom of substrate **202** may be grounded. Current leakage measurement performed through ohmic contact **223** on epitaxial layer **204** at an intermediate stage, such as those illustrated in FIGS. **10A** and **10B**, may serve as a quality control measurement similarly to the measurement in FIGS. **4C** and **5C**. If the device does not pass a current leakage measurement quality control test at an intermediate stage, such as those illustrated in FIGS. **10A** and **10B**, it may be discarded. Also other devices fabricated on substrates of the same batch may be discarded. If the device does pass a current leakage measurement quality control test at an intermediate stage, such as those illustrated in FIGS. **10A** and **10B**, a next fabrication stage may be performed. For example, for the stage illustrated in FIG. **10A**, such next stage may be forming metallization contacts M1.

(48) In one aspect of the present disclosure, a method of fabricating a device includes forming a plurality of structures, such that each structure of the plurality includes a substrate and an epitaxial

layer on the substrate. The epitaxial layer and the substrate have a lattice mismatch. The method further includes forming an electrical contact on the epitaxial layer of a selected structure of the plurality of structures and performing a current leakage measurement quality control test for the selected structure of the plurality of structures through the electrical contact. The method also includes forming a device on each of the remaining structures of the plurality of structures if the selected structure passed the leakage measurement quality control test or discarding each of the remaining structures of the plurality of structures if the selected structure did not pass the leakage measurement quality control test.

(49) In another aspect of the present disclosure, a method of forming a device includes epitaxially growing an epitaxial layer on each substrate of a batch of substrates. The epitaxial layer and the substrate have a lattice mismatch. The method further involves forming an ohmic contact on the epitaxial layer of a selected substrate of the batch; applying a voltage between the ohmic contact and a surface of the selected substrate, which is opposite to the epitaxial layer, and measuring an electrical current leakage between the epitaxial layer and the substrate. The method further involves forming a device on each remaining substrate of the batch if the measured leakage for the selected substrate does not exceed a threshold current leakage value or discarding each of the remaining substrates of the batch if the measured leakage of the electrical current for the selected substrate exceeds a threshold current leakage value. The device has an active region having a surface area which corresponds to a surface area of the ohmic contact on the epitaxial layer of the selected substrate.

(50) In yet another aspect of the present disclosure, a semiconductor device that includes a substrate and an epitaxial layer on the substrate. The epitaxial layer and the substrate have a lattice mismatch. The device includes a device region on a first part of a surface of the epitaxial layer, the device region comprising a plurality of transistors. The device also includes a seal ring region on a second part of the surface of the epitaxial layer. The seal ring region includes an ohmic contact on the epitaxial layer.

(51) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method of fabricating a device, comprising: forming a plurality of structures, each structure of said plurality consists of a substrate and an epitaxial layer on a top surface of the substrate, the epitaxial layer and the substrate have a lattice mismatch; forming an electrical contact on the epitaxial layer of a selected structure of said plurality to form a test structure, wherein the formed electrical contact is the only electrical contact of the formed test structure; performing a current leakage quality control test on the formed test structure, wherein said performing the current leakage quality control test comprises applying a voltage between the electrical contact and a bottom surface of the substrate of the test structure and measuring a current leakage between the epitaxial layer and the substrate of the test structure; and forming a device on each of the remaining structures of said plurality of structures if the measured leakage current for the test structure does not exceed a threshold leakage current value discarding each of the remaining structures of said plurality of structures if the measured leakage current of the test structure is greater than the threshold leakage current value test.

2. The method of claim 1, wherein the substrate comprises a Group IV semiconductor and the epitaxial layer comprises a Group III-V semiconductor.
3. The method of claim 1, wherein the substrate comprises silicon and the epitaxial layer is an epitaxial gallium nitride layer.
4. The method of claim 1, wherein the forming the electrical contact comprises forming a patterned metal layer on the epitaxial layer.
5. The method of claim 4, wherein the forming the electrical contact further comprises depositing an insulating layer over the patterned metal and etching the insulating layer to expose a surface of the patterned metal layer.
6. The method of claim 1, wherein said forming the device comprises forming a device region comprising a plurality of transistors.
7. The method of claim 6, wherein said forming the device further comprises forming a seal ring region surrounding the device region, wherein forming the seal ring region comprises forming an ohmic contact on the epitaxial layer of the structure.
8. The method of claim 7, further comprising performing a leakage current measurement through the ohmic contact of the seal ring region.
9. The method of claim 1, wherein the plurality of structures includes at least 8 structures.
10. The method of claim 1, wherein the epitaxial layer comprises a plurality of epitaxial sublayers.
11. The method of claim 5, wherein the test structure consists of the patterned metal layer, the epitaxial layer, the substrate and an unetched portion of the insulating layer around an edge of the patterned metal layer.
12. The method of claim 5, wherein the test structure consists of the patterned metal layer, the epitaxial layer, the substrate and an unetched portion of the oxide layer around an edge of the patterned metal layer.
13. A method of forming a device comprising: epitaxially growing an epitaxial layer on each substrate of a batch of substrates, wherein the epitaxial layer and the substrate have a lattice mismatch; forming an ohmic contact on the epitaxial layer of a selected substrate of the batch to form a test structure, wherein the formed ohmic contact is the only electrical contact of the test structure; applying a voltage between the ohmic contact and a surface of the substrate of the test structure, which surface is opposite to the epitaxial layer of the test structure, and measuring an electrical current leakage between the epitaxial layer and the substrate of the test structure; and forming a device on each remaining substrate of the batch if the measured leakage for the test structure does not exceed a threshold current leakage value or discarding each of the remaining substrates of the batch if the measured leakage of the electrical current for the test structure exceeds a threshold current leakage value, wherein the device has an active region having a surface area which corresponds to a surface area of the ohmic contact on the epitaxial layer of the test structure.
14. The method of claim 13, wherein the substrate is a Group IV semiconductor substrate and the epitaxial layer comprises a Group III-V semiconductor.
15. The method of claim 13, wherein the substrate is a silicon substrate and the epitaxial layer comprises GaN.
16. The method of claim 13, wherein said forming the device comprises forming a device region comprising a plurality of transistors.
17. The method of claim 13, wherein the forming the ohmic contact comprises forming a metal layer on the epitaxial layer and patterning the metal layer.
18. The method of claim 17, wherein the forming the ohmic contact further comprises depositing an oxide layer over the patterned metal layer and etching the oxide layer to expose a surface of the patterned metal layer.
19. A method of fabricating a device, comprising: forming a plurality of structures, each structure of said plurality of structures comprising a substrate and an epitaxial layer on the substrate, the epitaxial layer and the substrate having a lattice mismatch; forming an electrical contact on the

epitaxial layer of a selected structure of said plurality of structures; performing a current leakage quality control test for the selected structure of said plurality of structures through said electrical contact; and forming a device on each of the remaining structures of said plurality of structures if the selected structure passed the current leakage quality control test or discarding each of the remaining structures of said plurality of structures if the selected structure did not pass the leakage measurement quality control test, wherein said forming the device comprises forming a device region comprising a plurality of transistors and forming a seal ring region surrounding the device region, wherein forming the seal ring region comprises forming an electrical contact on the epitaxial layer of the structure.

20. The method of claim 19, further comprising performing a leakage current measurement through the electrical contact of the seal ring region.
